idea: use binary search as a 'global estimator'? → not thinking guessed → not efficient ⇒ no bounds

[0x1f]

cmp ⇒ ZF = 1 → rax = 0 (why?)
→ rax ≠ {0,1} ⇒ rax < 0?

Collatz conjecture: $f(n) \longrightarrow 1$ $\forall n \in \mathbb{N}$ (or $\forall n \in \mathbb{Z}^+$)

$$f(n) = \begin{cases} n/2 & n \equiv 0 \pmod 2 \\ 3n+1 & n \equiv 1 \pmod 2 \end{cases}$$

msb = 2 ← 0-indexed

ex) 0100 shr 2 → 0001 would be?

```
.loop:
    bsf     rcx, rax
    shr     rax, cl
    cmp     rax, 1
    je      .exit_loop
    lea     rax, [rax + 2*rax + 1]
    jmp     .loop
.exit_loop:
```

cl ∈ [0, 255]

jnz ?

3*rax + 1

value is negative

je

bsf: ∃ msb ∈ rax ⇒ ZF := 1
ie rax ≠ 0

1000 ⇒ 3
0100 ⇒ 2

8 rcx = 0-indexed position

lea: lea dest, [base + scale*index + offset]

Wishy-guess: memory-chunk search (acceptable)

∃? N → N other stuff → ∃? → ∃ Y

cl

N?

∀ exit

cl = 255

msb @

[0x1f]

is rax > max(cl)?

Y

rax = 00(MSB)0...

ex) 00000001 ⇒ msb @ [7]

01000001 → msb @ [6]

0001000001 → msb @ [4]